

Simulation-Native Worlds

Toward Material-Based Interactive Reality

1. Abstract

Modern games have achieved unprecedented levels of visual fidelity, yet their worlds remain fundamentally surface-based constructions composed of polygons, textures, scripted interactions, and approximated physics.

> Games evolved visual fidelity faster than simulation fidelity.

Despite increasingly realistic rendering, most interactive environments remain structurally non-physical: destruction is preauthored, materials are superficial, and object behavior is constrained by predefined systems rather than emerging from the world itself.

This paper proposes the concept of **simulation-native material architecture** — an alternative model in which game worlds are represented not as visual shells, but as material systems defined by structure, physical properties, and dynamic interaction. In such a model, physics becomes the foundation of world representation rather than a secondary overlay, enabling environments that behave as coherent physical systems rather than scripted audiovisual approximations.

2. Introduction

Modern games strive for realism, but this realism is primarily visual rather than systemic.

Game worlds:

- * look realistic,
- * but do not behave like real environments.

> Realism in modern games is still largely a property of rendering, not behavior.

Even the most advanced open-world environments remain fundamentally non-physical. Objects rarely possess internal structure, materials do not exist as true simulation entities, and interaction systems are constrained by predefined logic and authored responses.

As a result, contemporary game worlds often simulate the appearance of reality more effectively than reality itself.

This paper explores the possibility of transitioning from polygon-based visual approximation toward **material-simulation reality** — a model in which physical properties become the basis of interactivity and systemic behavior.

3. Limits of Polygon-Based Worlds

Polygon Worlds as Surface Representation

Traditional game environments are primarily built from:

- * polygon meshes,
- * textures,
- * shaders,
- * collision shells,
- * authored animation systems.

Objects appear material, but are not materially defined.

A wooden wall is visually identified as wood, yet internally it is often no different from stone, plastic, or metal beyond predefined gameplay variables.

This creates the illusion of matter, but not matter itself.

Physics as a Secondary Layer

In most modern engines, physical behavior is layered on top of visual representation rather than constituting the world itself.

Physics systems:

- * approximate interaction,
- * operate independently from object structure,
- * function as isolated subsystems.

> Physics is typically a plugin to the world, not the foundation of it.

As a result, physicality remains constrained, selective, and heavily simplified.

Interaction as Scripted Behavior

Most environmental interactions are predefined:

- * destruction patterns are authored manually,
- * reactions are animation-driven,
- * object states are limited to predefined transitions.

The player often interacts not with a simulated world, but with a collection of scripted triggers and handcrafted responses.

Interaction complexity is therefore limited not by systemic possibility, but by production scope.

Visually Complex, Structurally Static Worlds

Modern games can achieve extraordinary visual scale and detail while remaining structurally static.

Most worlds lack:

- * internal material continuity,
- * structural stress,
- * dynamic deformation,
- * persistent physical consequences.

Environments may appear alive visually while remaining fundamentally inert as systems.

> The world remains decoration rather than simulation.

4. Simulation-Native Reality

Central Idea

> Game worlds represented through material simulation rather than polygon illusion.

Simulation-native reality proposes a fundamentally different representation model in which materials become the primary substrate of interactive environments.

In such architectures, objects are not defined primarily by surfaces, but by:

- * composition,
- * structure,
- * density,
- * strength,
- * deformation properties,
- * thermal behavior,
- * reactions to force and stress.

The world is no longer treated as a rendered approximation of matter, but as a material system capable of generating behavior dynamically.

Material-Based Objects

In a simulation-native architecture, objects are represented through:

- * material composition,
- * structural relationships,
- * internal connectivity,
- * distribution of mass,
- * mechanical properties.

Objects no longer merely resemble wood, metal, or concrete visually.

Within the simulation, they behave according to the systemic properties of those materials.

Emergent Physical Interaction

Interaction emerges from:

- * environmental conditions,
- * force distribution,
- * material resistance,
- * structural instability,
- * accumulated stress.

Behavior is not explicitly scripted.

Instead, it becomes the consequence of interacting systems.

Destruction is therefore not a predefined animation event, but a physical outcome emerging from the state of the environment itself.

Structural Reality

Simulation-native worlds possess:

- * structural continuity,
- * points of weakness,
- * dynamic stability,
- * physical failure conditions.

Buildings, terrain, and objects behave as interconnected structures rather than independent mesh entities.

As a result, environments become reactive systems rather than static visual compositions.

Unified Simulation Layer

In traditional game architecture:

- * rendering,
- * physics,
- * gameplay logic,
- * destruction systems

are often separated into loosely connected layers.

Simulation-native architecture instead treats physical simulation as the foundational layer of world representation.

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```text id="kxj75h"
material structure
 ↓
physical simulation
 ↓
emergent interaction
 ↓
visual reconstruction
```
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Rendering visualizes simulation.

Gameplay emerges from simulation.

AI assists and optimizes simulation.

The world is no longer authored primarily as geometry, but as behavior.

5. The Role of AI

Within simulation-native architectures, AI serves an enabling rather than central role.

Its purpose is not to replace physical systems, but to make large-scale simulation computationally feasible.

AI may:

- * optimize simulation fidelity,
- * approximate expensive calculations,
- * reconstruct visual detail,
- * manage scalability,
- * predict system behavior,
- * reduce computational overhead.

AI therefore functions as an adaptive simulation management layer rather than as the core reality model itself.

- > AI does not replace physics.
- > It enables simulation to exist at scale.

6. Implications for Interactive Systems

Simulation-native worlds potentially enable entirely new categories of interactivity:

- * emergent gameplay,
- * reactive environments,
- * persistent systemic consequences,
- * dynamic structural interaction,
- * autonomous environmental evolution,
- * simulation-driven problem solving.

Game design shifts:

- * from scripted events → to systemic emergence,
- * from predefined mechanics → to material interaction,
- * from static environments → to reactive worlds.

As simulation depth increases, gameplay becomes less dependent on handcrafted interaction trees and more dependent on the dynamic state of the world itself.

This may fundamentally change how interactive experiences are designed, authored, and played.

7. Challenges and Open Questions

Despite its potential, simulation-native architecture introduces major challenges.

Computational Cost

Large-scale material simulation is computationally expensive and difficult to execute in real time.

Scalability

Maintaining coherent simulation across large environments remains a significant technical problem.

Controllability

Highly emergent systems may become difficult to balance, predict, or design around.

Simulation vs Gameplay

Perfect realism does not necessarily produce compelling gameplay.

A critical challenge lies in balancing:

- * systemic authenticity,
- * player readability,
- * interaction clarity,
- * gameplay pacing.

Production Paradigm Shift

Simulation-native worlds may require entirely new production pipelines, tools, and design methodologies beyond traditional content-driven development.

8. Conclusion

The transition from polygon-based architectures toward simulation-native reality represents more than a technical evolution.

It suggests a fundamental shift in how interactive worlds are represented, constructed, and experienced.

For decades, game development has primarily advanced through increasing visual fidelity. Yet visual realism alone cannot create worlds that behave as coherent systems.

Simulation-native architecture proposes an alternative trajectory:

- * from surfaces → to materials,
- * from scripted interaction → to systemic emergence,
- * from static environments → to reactive worlds,
- * from visual approximation → to simulation authenticity.

> The next evolution of game worlds may emerge not from rendering better illusions, but from simulating reality itself.